Supplemental Material for "Roton-like Reconstruction of the Emergent Photon in Quantum Spin Ice"

Zhengbang Zhou[1] and Yong Baek Kim[1]
[1]*Department of Physics, University of Toronto, Toronto, Ontario M5S 1A7, Canada*

THEORETICAL RESPONSE AND RELATION TO NEUTRON SCATTERING

The quantity calculated in the Letter is the zero-temperature, sublattice-summed longitudinal pseudospin correlation function

$$S^{zz}(\boldsymbol{q},\omega) = \frac{1}{N}\sum_{\mu,n} \left|\langle n|S^z_\mu(\boldsymbol{q})|0\rangle\right|^2 \delta(\omega-\omega_n), \tag{1}$$

where

$$S^z_\mu(\boldsymbol{q}) = \sum_{i\in\mu} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} S^z_i \tag{2}$$

and μ runs over the four pyrochlore sublattices.

Inside the spin-ice manifold, S^z_i is the component of the pseudospin along the local $\langle 111\rangle$ easy axis and equals the emergent electric field,

$$E_i = S^z_i. \tag{3}$$

Every quoted spectral fraction is normalized at fixed momentum:

$$Z_n(\boldsymbol{q}) = \frac{\sum_\mu |\langle n|S^z_\mu(\boldsymbol{q})|0\rangle|^2}{\sum_{m,\mu} |\langle m|S^z_\mu(\boldsymbol{q})|0\rangle|^2}. \tag{4}$$

Equation (1) is not by itself the laboratory neutron cross section. A material-specific magnetic cross section contains the ionic form factor, the g tensor, the rotation between the four local $\langle 111\rangle$ frames and the laboratory frame, the neutron polarization tensor

$$\delta_{\alpha\beta} - \hat{q}_\alpha\hat{q}_\beta, \tag{5}$$

and coherent interference between different sublattices.

Schematically,

$$I(\boldsymbol{q},\omega) \propto \sum_{\mu\nu} \mathcal{F}_{\mu\nu}(\boldsymbol{q}) S^{zz}_{\mu\nu}(\boldsymbol{q},\omega), \tag{6}$$

where

$$S^{zz}_{\mu\nu}(\boldsymbol{q},\omega) = \sum_n \langle 0|S^z_\mu(-\boldsymbol{q})|n\rangle\langle n|S^z_\nu(\boldsymbol{q})|0\rangle\delta(\omega-\omega_n). \tag{7}$$

The 50–54% low-mode fractions quoted in the Letter therefore refer strictly to the sublattice-summed pseudospin response. They are not universal neutron-intensity fractions. A material-specific prediction requires applying the coherent projection appropriate to the material's local axes and g tensor.

CONTRACTIBLE RING TERMS AND FINITE CLUSTERS

The Hamiltonian is

$$H = -K\sum_p \left(W_p + W_p^\dagger\right), \qquad W_p = S^+_1 S^-_2 S^+_3 S^-_4 S^+_5 S^-_6. \tag{8}$$

On a periodic finite cluster, six-site closed paths come in two classes: elementary contractible hexagons corresponding to plaquettes of the infinite pyrochlore lattice, and loops that close only by winding around the torus. The latter are not local plaquettes of the thermodynamic Hamiltonian. We therefore compute the winding vector of every six-cycle and retain only zero-winding elementary hexagons.

For every cluster used in the Letter, the retained Hamiltonian has the full lattice density of elementary rings.

The principal finite clusters are

	$2 \times 2 \times 3$	$2 \times 2 \times 4$	$2 \times 3 \times 3$	$2 \times 3 \times 4$
sites	48	64	72	96
ice configurations	87 546	2 756 394	12 846 186	2 007 260 562
spectral sector dimension	9 104	249 180	973 008	146 758 352

The commonly used 16- and 32-site pyrochlore clusters are pathological for the present dynamical question. On the 32-site cell every configuration in the connected zero-flux sector has exactly eight flippable hexagons. The equal-amplitude state is consequently an exact eigenstate and the spectrum becomes artificially integer-like. On the 16-site cluster only six transport sectors possess nontrivial ring dynamics. We therefore begin with 48 sites.

Varying cluster shape at fixed spin number is also not a clean option. At these sizes, alternative thin tori containing an L-type momentum lose a substantial fraction of the contractible elementary rings. For example, the $1 \times 3 \times 4$ 48-site cell contains only 24 contractible hexagons. Keeping only those terms produces a Hamiltonian with half the ring density and therefore a different microscopic model. The finite-size analysis consequently uses the sequence of admissible clusters listed above.

WINDING SECTORS

The ring Hamiltonian conserves topological winding polarization. Configurations are assigned an exact integer key derived from

$$\boldsymbol{P} = \sum_i \left(n_i - \frac{1}{2}\right) \boldsymbol{r}_i. \tag{9}$$

On the 48-site cluster all winding sectors can be enumerated explicitly. The global ground state lies in the sector of dimension 9104 used for complete diagonalization.

On 72 sites the zero-polarization sector is empty. The ground states instead form a degenerate $\pm\boldsymbol{P}$ pair of nonzero-polarization sectors, each of dimension

$$973\,008 \tag{10}$$

and energy

$$E_0 = -14.80971536K. \tag{11}$$

The identification is verified by independent diagonalization of the largest candidate sectors.

ENUMERATION AND LANCZOS SPECTROSCOPY

Ice configurations are enumerated exactly by assigning spins sequentially and backtracking whenever a tetrahedron would acquire three spins of either sign. For every elementary hexagon, a bit mask and its two alternating flippable patterns identify the states connected by the ring Hamiltonian.

On 48 sites the 9104-dimensional sector is diagonalized completely with a dense symmetric eigensolver. All eigenvalues and eigenvectors are therefore available.

At 64 sites the sector dimension is 249 180. The dynamical response is obtained through Lanczos continued fractions seeded independently by

$$S^z_\mu(\boldsymbol{q})|0\rangle. \tag{12}$$

The 64-site calculation uses full reorthogonalization.

Going beyond 64 spins requires replacing the single-word bit representation. For 72 and 96 sites, each configuration is represented by a pair of 64-bit words. All masking, flipping, sorting, and exact lookup operations are implemented on the word pair.

The two-word pipeline is validated on the 48-site cluster, where it reproduces

$$E_0 = -10.55601176K, \tag{13}$$
$$S(L) = 0.37036, \tag{14}$$

and the principal L energies

$$1.0235K, \qquad 1.7303K \tag{15}$$

exactly relative to the single-word implementation.

On 96 sites the complete ice manifold contains

$$2\,007\,260\,562 \tag{16}$$

configurations. It is enumerated using a vectorized frontier algorithm that imposes one tetrahedron constraint at a time and branches the frontier into independent seeds. The spectral sector contains

$$146\,758\,352 \tag{17}$$

basis states and approximately

$$2.7 \times 10^9 \tag{18}$$

nonzero Hamiltonian matrix elements.

The continued-fraction calculation uses real Lanczos separately on the real and imaginary parts of the probe. This is exact for a real symmetric Hamiltonian and reduces Krylov storage. On the 72-site cluster, 120 real iterations reproduce the 250-iteration complex calculation to every printed digit.

Degenerate eigenvalues are grouped within

$$10^{-8}K \tag{19}$$

and their spectral weights summed before branch weights are quoted. Only the total weight of a degenerate multiplet is basis invariant.

Momentum labels are certified by explicit translation projectors rather than inferred from an ordering of the numerical eigenstates.

Three exact gates are applied throughout:

1. the plaquette-energy identity

$$\sum_p \langle W_p + W_p^\dagger \rangle = -\frac{E_0}{K}, \tag{20}$$

2. agreement of the spectral first moment with the exact double-commutator result, and

3. the gauge identity

$$Cd_0 = 0 \tag{21}$$

for the lattice curl.

GAUSSIAN LATTICE ELECTRODYNAMICS

Every comparison with a "Gaussian photon" is made using the lattice curl on the same finite torus as the microscopic calculation.

Write

$$B_p = \sum_\ell C_{p\ell} A_\ell, \tag{22}$$

where C is the alternating lattice curl. Expanding the compact magnetic energy to quadratic order and introducing the electric stiffness gives

$$H_G = \frac{U}{2} \sum_\ell E_\ell^2 + K \sum_p B_p^2, \qquad [A_\ell, E_{\ell'}] = i\delta_{\ell\ell'}. \tag{23}$$

At fixed momentum, the normal-mode frequencies are determined by the nonzero singular values of the curl block,

$$\omega_\lambda(\boldsymbol{q}) = \sqrt{2UK}\,\sigma_\lambda(\boldsymbol{q}). \tag{24}$$

The construction satisfies three checks:

1. gauge invariance,

$$Cd_0 = 0 \tag{25}$$

   to machine precision;

2. exactly two nonzero singular values for every $\boldsymbol{q} \neq \Gamma$ and none at Γ;

3. orthonormality and completeness of the Bloch basis.

The pure ring Hamiltonian contains no independent bare value of U, so one overall scale must be inferred from the interacting spectrum. A single linear least-squares fit gives

$$\sqrt{2UK} = 0.5868K. \tag{26}$$

At L this gives

$$\omega_G(L) = 2.03K. \tag{27}$$

The fitted curve is used only to quantify the magnitude of the reconstruction and to compare momentum dependence. The scale-free branch ratio and spectral-weight fractions quoted in the Letter do not depend on this fit.

BASIS-INVARIANT TRANSVERSE-CHANNEL RESOLUTION

At each momentum the electric-field probe spans the four-dimensional sublattice space. The lattice curl has rank two, leaving a two-dimensional transverse physical subspace. The residual weight in the remaining longitudinal directions is less than

$$10^{-31} \tag{28}$$

relative to $S(\boldsymbol{q})$.

The two Gaussian singular values are exactly degenerate, so an individual transverse basis is defined only up to a rotation. Statements assigning a level to a particular arbitrarily chosen Gaussian polarization are therefore not invariant.

For a level of multiplicity m, let A_n be the $m \times 2$ matrix of amplitudes in any orthonormal transverse basis. We define

$$M_n = A_n^\dagger A_n. \tag{29}$$

The polarization purity is

$$\mathcal{P}_n = \frac{\operatorname{tr} M_n^2}{(\operatorname{tr} M_n)^2}, \tag{30}$$

and the normalized overlap between two levels is

$$\mathcal{O}_{AB} = \frac{\mathrm{tr}(M_A M_B)}{\mathrm{tr}M_A\,\mathrm{tr}M_B}. \tag{31}$$

Both quantities are invariant under rotations of the transverse basis.

Representative 48-site results are

$\boldsymbol{q}$	ω_1/K	ω_2/K	splitting	overlap
$(\frac{1}{3},\frac{1}{3},\frac{1}{3})$	1.895	1.980	0.044	2×10^{-28}
$(\frac{1}{3},\frac{1}{3},\frac{2}{3})$	1.744	2.069	0.171	10^{-29}
$(\frac{1}{6},\frac{1}{6},\frac{5}{6})$	2.533	2.687	0.059	4×10^{-2}
L	1.024	1.730	0.513	3×10^{-3}
X	2.235	3.423	0.420	10^{-31}

The two brightest levels are nearly orthogonal directions within the two-dimensional transverse probe space. We use this result to speak operationally of two transverse S^z-active branches. The polarization geometry by itself is not taken as proof of an adiabatic quasiparticle ancestry.

EXACT FIRST MOMENT AND FEYNMAN RATIO

The first spectral moment is

$$f_1(\boldsymbol{q}) = \frac{1}{2N}\sum_\mu \langle 0|[S^z_\mu(-\boldsymbol{q}),[H,S^z_\mu(\boldsymbol{q})]]|0\rangle. \tag{32}$$

For the ring Hamiltonian this reduces exactly to

$$f_1(\boldsymbol{q}) = \frac{K}{2N}\sum_p \langle W_p + W_p^\dagger\rangle \sum_\mu \left|g_p^\mu(\boldsymbol{q})\right|^2. \tag{33}$$

The zeroth moment is

$$S(\boldsymbol{q}) = \int d\omega\, S^{zz}(\boldsymbol{q},\omega), \tag{34}$$

so the normalized state proportional to $S^z(\boldsymbol{q})|0\rangle$ has variational energy

$$\omega_{\mathrm{SMA}}(\boldsymbol{q}) = \frac{f_1(\boldsymbol{q})}{S(\boldsymbol{q})}. \tag{35}$$

This ratio is identically the spectral centroid. Its value after a spectrum has been calculated is therefore not independent evidence for a low pole. The nontrivial tests used in the Letter are instead the predictions made using the ground-state ratio before the corresponding excited states are calculated.

GEOMETRY AND THE INTERACTING ENHANCEMENT OF $S(L)$

The oscillator strength has an especially transparent geometric structure. If a hexagon lies inside a wavefront of the electric-field oscillation, the alternating loop integral vanishes identically and the plaquette contributes nothing to Eq. (33).

At

$$L \parallel [111], \tag{36}$$

one of the four hexagon families lies in the wavefronts. On the 48-site cluster the family-resolved geometric factors are

$$(96, 96, 96, 0)_L, \tag{37}$$

compared with

$$(96, 96, 96, 96)_X. \tag{38}$$

The reduction in the pure geometric factor is therefore exactly 3/4. The reduction of f_1 itself is not exactly 3/4 because the ground-state plaquette coherences are not perfectly uniform on an anisotropic cluster. Numerically,

$$\frac{f_1(L)}{f_1(X)} = \frac{0.684}{0.880} = 0.777. \tag{39}$$

The geometric cancellation alone does not generate the anomaly. At the shortest sampled momentum along [111], one family is also exactly dark, but the dominant level remains near the Gaussian photon.

The second ingredient is the interacting equal-time structure factor. On 48 sites,

$$S(L) = 0.370 \tag{40}$$

for the quantum ground state, compared with

$$S_{\mathrm{EA}}(L) = 0.297 \tag{41}$$

for the equal-amplitude superposition over the same 9104 configurations.

A classical loop-update calculation on a 2048-site ice manifold gives

$$S(L) = 0.2488(67), \tag{42}$$
$$S(X) = 0.2538(73), \tag{43}$$
$$S(W) = 0.2475(52), \tag{44}$$
$$S(K) = 0.2467(46). \tag{45}$$

Within statistical precision the classical zone-boundary structure factor is essentially flat. The enhancement near L is therefore generated predominantly by the interacting quantum amplitudes at the physical ring point.

The 72-site soft-valley momentum

$$\boldsymbol{q}_v = \left(\frac{1}{6}, \frac{1}{2}, \frac{1}{2}\right) \tag{46}$$

has geometric family weights

$$(36, 144, 144, 108), \tag{47}$$

so no family is exactly dark. Nevertheless its enhanced structure factor produces an f_1/S within 0.9% of the L value, and the two reconstructed levels are within 0.7% of one another.

ROKHSAR–KIVELSON DEFORMATION

We use the standard RK extension

$$H(V) = -K \sum_p \left(W_p + W_p^\dagger\right) + V \sum_p n_p^{\mathrm{flip}}, \tag{48}$$

where n_p^{flip} projects onto a flippable alternating hexagon.

Our sign convention is such that

$$V > 0 \tag{49}$$

penalizes flippability,

$$V < 0 \tag{50}$$

rewards it, and the exactly solvable RK point is at

$$V = +K. \tag{51}$$

For state characterization we use only the derivative at the physical ring point $V = 0$. Hellmann–Feynman gives

$$\frac{d\omega_n}{dV} = \langle n\,|N_{\text{flip}}|\,n\rangle - \langle 0\,|N_{\text{flip}}|\,0\rangle\,, \qquad N_{\text{flip}} = \sum_p n_p^{\text{flip}}. \tag{52}$$

Thus $d\omega_n/dV$ is an exact state-resolved measure of the difference in the number of flippable plaquettes between an excitation and the ground state.

At 48 sites the reconstructed L level has

$$\frac{d\omega_{\text{low}}}{dV} = +0.339. \tag{53}$$

The upper L level responds with the opposite sign.

Across the four finite sizes the reconstructed levels have

$$+0.339, \quad +0.338, \quad +0.134, \quad +0.259. \tag{54}$$

The sign remains positive, but the magnitude is not size stable. We therefore do not use the census as the primary discriminator.

Exact census sum rule

The operator N_{flip} is diagonal in the configuration basis. Every flippable hexagon generates exactly one off-diagonal ring-connected state. Consequently

$$\operatorname{Tr} N_{\text{flip}} = \operatorname{nnz}(H), \tag{55}$$

where $\operatorname{nnz}(H)$ denotes the number of nonzero off-diagonal matrix elements, with the same counting convention.

Averaging Eq. (52) over the complete spectrum of a sector of dimension D gives

$$\frac{1}{D}\sum_n \frac{d\omega_n}{dV} = \frac{\operatorname{nnz}(H)}{D} - \langle 0|N_{\text{flip}}|0\rangle. \tag{56}$$

On the 48-site sector,

$$\frac{1}{D}\sum_n \frac{d\omega_n}{dV} = -0.9139, \tag{57}$$

and the direct average over all 9104 eigenstates reproduces this value to 10^{-14}.

Excitations therefore lose flippability on average. There is no sign theorem for an individual eigenstate. Of the 9104 eigenstates,

$$58 \tag{58}$$

have positive census and

$$20 \tag{59}$$

exceed $+0.20$. The reconstructed L state at $+0.339$ lies in the extreme positive tail but is not unique.

The cleaner census result is the within-cluster comparison. The 64- and 96-site clusters each contain two inequivalent L-type classes at identical $|\boldsymbol{q}|$ and identical Gaussian energy. Their censuses are

$$64: \quad +0.338 \quad \text{versus} \quad -0.166, \tag{60}$$

$$96: \quad +0.259 \quad \text{versus} \quad -0.02,\ -0.09. \tag{61}$$

The classes that reconstruct and those that do not therefore respond with opposite signs to the same microscopic flippability probe.

Finite RK trajectory

For completeness we also follow the reconstructed level over a finite range of V. Biasing toward more flippable configurations, $V < 0$, increases $S(L)$ and softens the low level from approximately

$$1.02K \tag{62}$$

to

$$0.73K \tag{63}$$

over the range studied, while $f_1(L)$ changes by less than one percent.

This evolution is consistent with the Feynman-ratio description: the softening accompanies an enhancement of the equal-time fluctuation while the oscillator strength is nearly unchanged. We do not interpret the deformation as establishing a temporal or causal ordering between ground- and excited-state quantities.

VARIATIONAL RELAXATION OF THE RECONSTRUCTED STATE

The sublattice-resolved probe vectors

$$S^z_\mu(L)|0\rangle \tag{64}$$

span the transverse electric-field trial space. The lowest energy in this space on the 48-site cluster is

$$1.450K. \tag{65}$$

A systematic variational improvement converges rapidly:

$$1.450 \to 1.074 \to 1.034 \to 1.026K, \tag{66}$$

compared with the exact eigenvalue

$$1.024K. \tag{67}$$

Thus the reconstructed state is reached in a small collective subspace generated from the physical probe.

The fractional recovery relative to the bare optimized electric-field state is

$$\frac{\omega_{\mathrm{bare}} - \omega_{\mathrm{exact}}}{\omega_{\mathrm{bare}}} = 29\% \tag{68}$$

on 48 sites. The corresponding values on 48, 64, and 72 sites are

$$29\%, \qquad 32\%, \qquad 28\%. \tag{69}$$

BASIS-INVARIANT PLAQUETTE DECOMPOSITION

Because the ring term is the entire nonconstant Hamiltonian in the ice manifold, every excitation energy decomposes exactly into plaquette contributions:

$$\omega_n = -K \sum_p \left[\langle n|W_p + W_p^\dagger|n\rangle - \langle 0|W_p + W_p^\dagger|0\rangle \right]. \tag{70}$$

Grouping plaquettes into the four $\langle 111\rangle$ orientation families gives a complete four-term decomposition.

The exact low level is degenerate. Expectation values in an arbitrary basis of this multiplet are therefore not physically invariant. We choose the normalized projection of the physical S^z probe onto the degenerate multiplet. This state is basis independent. An alternative alignment to the optimized single-mode state gives per-family values identical to four decimals.

The resulting decomposition is

	48	64	72
single-mode energy	1.450	1.412	1.481
exact low level	1.024	0.954	1.068
recovery	29%	32%	28%
relief, family 1	+0.080	+0.040	+0.276
relief, family 2	−0.064	−0.036	+0.002
relief, family 3	+0.088*	+0.227	+0.002
relief, family 4	+0.322	+0.227	+0.133*
dark-family share	21%	9%	32%

where $\star$ denotes the geometrically dark family for the corresponding cluster.

The 28–32% relaxation is stable across the three sizes. It is strongly family selective, but it is not concentrated in the geometrically dark family.

An earlier analysis obtained a 71% dark-family contribution by evaluating an arbitrary member of the degenerate multiplet. That result is basis dependent and is withdrawn. The corrected conclusion is a robust, family-selective reorganization of ring coherence with a dark-family contribution of 9–32%.

FEW-PHOTON KINEMATICS

Global spin reversal is an exact symmetry of the ring Hamiltonian. For an even ground state the linear S^z probe is odd and therefore couples only to odd-parity eigenstates. In gauge language this is charge conjugation, under which a state containing n weakly defined photons has parity

$$(-1)^n. \tag{71}$$

Parity excludes even photon number but does not exclude three-photon states. We therefore minimize the energy of finite-volume Gaussian photon partitions with total momentum L.

On the 48-site torus the resulting thresholds are

$$\omega_L^{(1)} = 2.03K, \tag{72}$$

$$\omega_L^{(2)} = 4.03K, \tag{73}$$

$$\omega_L^{(3)} = 5.55K. \tag{74}$$

The reconstructed level lies at

$$1.02K, \tag{75}$$

more than a factor of five below the weakly renormalized Gaussian three-photon threshold.

This rules out an interpretation in terms of a few weakly interacting Gaussian photons on the finite clusters. Because the Letter itself establishes order-one interactions at the zone boundary, the calculation is not a theorem excluding multiparticle states formed from strongly renormalized constituents in the thermodynamic theory.

OFF-POLE SPECTRAL WEIGHT AND COMPLEMENTARY PROBES

Beyond the two dominant transverse branches,

$$16\%\text{–}38\% \tag{76}$$

of the 48-site S^z spectral weight resides in higher levels absent from the Gaussian one-photon theory. The off-pole fraction is smallest at the shortest accessible momenta, consistent with recovery of the infrared photon.

Spin-reversal parity gives a clean separation of probe channels. Numerically, every even state carries linear S^z weight below

$$10^{-28}. \tag{77}$$

Bilinear operators can instead couple to the even sector, including two-particle channels, and can be accessed by probes such as Raman scattering or ultrasound [1, 2]. Magnetization probes such as neutron scattering and stray-field noise belong to the odd channel [3].

GREEN'S-FUNCTION MONTE CARLO

The ring Hamiltonian is stoquastic in the ice basis: its diagonal matrix elements vanish and every nonzero off-diagonal element is $-K$. A power method with

$$\Lambda - H \tag{78}$$

therefore converges to the ground state with nonnegative amplitudes.

We use the equal-amplitude trial state

$$|\Psi_T\rangle = \sum_x |x\rangle. \tag{79}$$

Population control is performed by comb reconfiguration at fixed walker number.

For diagonal observables such as $S(\boldsymbol{q})$, the mixed distribution $\Psi_0\Psi_T$ is converted to the pure distribution $|\Psi_0|^2$ by forward walking. An observable measured at time t is weighted by the number of descendants retained by the walker after a projection time T.

Convergence is benchmarked on the exactly diagonalized 48-site cluster using

$$256\text{–}2048 \tag{80}$$

walkers and

$$T = 20\text{–}160. \tag{81}$$

At $T = 20$, the forward-walking estimate of $S(L)$ is biased low by approximately 0.7%. For

$$T \geq 40 \tag{82}$$

the values fluctuate around the exact result

$$S(L) = 0.370365 \tag{83}$$

within statistical errors. Production runs use

$$T = 80. \tag{84}$$

The mixed-estimator energies exhibit a residual population-control bias of approximately

$$1.5 \times 10^{-4} \tag{85}$$

in relative units, independent of walker number over the tested range. This is two orders of magnitude below the energy differences used in the analysis.

Initialization requires care because sublattice-uniform ice configurations are frozen under the ring dynamics. Seeds are randomized using worm-constructed loop flips that first reduce the integer polarization and subsequently preserve the minimal reachable polarization exactly.

Every production calculation is gated against exact diagonalization. On 48 sites,

$$E_0 = -10.556012K \tag{86}$$

is reproduced within 0.5σ, and

$$S(L) = 0.370365 \tag{87}$$

is reproduced to every printed digit.

On 72 sites the method reproduces

$$E_0/N = -0.205690K \tag{88}$$

within 1.8σ in the correct nonzero-polarization ground sector.

Production calculations use 1024 walkers and 30 000–60 000 propagation steps. Statistical uncertainties are estimated from 16 bins.

On cubic supercells every hexagon is symmetry equivalent. The exact identity

$$\sum_p \langle W_p + W_p^\dagger \rangle = -\frac{E_0}{K} \tag{89}$$

therefore fixes the uniform plaquette coherence directly from the ground-state energy. The remaining momentum dependence of f_1 is exact lattice geometry.

The complete momentum grids contain 216 momenta on the 864-site 6^3 cell and 512 momenta on the 2048-site 8^3 cell.

LARGE-SYSTEM CENTROID LANDSCAPE

At the principal zone-boundary momenta the Feynman ratios are

	864	2048
L	2.142(10)	2.179(19)
X	2.660(10)	2.713(21)
W	—	2.781(10)
K	—	2.718(20)
U	—	2.736(11)

The L value shifts by only 1.7% from 864 to 2048 sites and remains 20–25% below the other principal zone-boundary momenta.

Along the radial [111] direction the centroid rises monotonically toward L. On the 8^3 cell the sequence is approximately

$$0.60, \quad 1.36, \quad 1.97, \quad 2.18 \tag{90}$$

at the allowed momenta

$$\frac{k}{8}(1,1,1). \tag{91}$$

Thus the centroid does not possess a helium-style radial dip. Instead, the soft structure is a minimum within the zone-boundary landscape.

Representative nearby values are

$$\frac{f_1}{S}\left(\frac{1}{4},\frac{1}{2},\frac{1}{2}\right) = 2.216(5), \tag{92}$$

$$\frac{f_1}{S}\left(\frac{3}{8},\frac{3}{8},\frac{5}{8}\right) = 2.265(5), \tag{93}$$

$$\frac{f_1}{S}\left(\frac{1}{4},\frac{1}{2},\frac{3}{4}\right) = 2.484(5). \tag{94}$$

The result is a shallow valley extending away from L rather than an isolated point.

EXPERIMENTAL INTERPRETATION

The calculation does not imply that every zone-boundary feature of a microscopic quantum-spin-ice material must coincide quantitatively with the low L level of the pure ring model. Longer ring exchanges, virtual spinons, further microscopic interactions, disorder, and the material-specific magnetic moment operator can modify the spectrum.

The calculation instead demonstrates that the minimal compact gauge Hamiltonian can itself redistribute a large fraction of the finite-momentum S^z spectral weight without leaving the deconfined ice manifold.

A useful experimental pattern to search for is therefore the following: a response that is dominated by a single photon-like feature near the zone center need not remain single-mode at the zone boundary. Along [111], substantial spectral weight may transfer into a lower feature while a weaker higher-energy feature remains near the continuation of the long-wavelength photon.

The exact intensity ratio is not universal. The physical neutron response coherently combines sublattice amplitudes before taking the squared modulus. The local easy axes and g tensor can therefore reweight the two branches differently and may enhance or suppress a particular feature in a given scattering geometry.

A finite cluster also produces discrete poles where the thermodynamic system may instead show a peak plus continuum. The robust experimental prediction of the minimal model is therefore *redistribution and reconstruction of finite-momentum spectral weight*, not a universal number of poles or a universal low/high intensity ratio.

SCOPE AND LIMITATIONS

The results establish the following directly:

1. a dominant reconstructed S^z level exists on all four finite clusters from 48 to 96 sites;
2. its weight fraction and scale-free branch ratio are stable across those clusters;
3. an RK flippability derivative provides an independent microscopic fingerprint of the reconstructing momentum class;
4. the ground-state Feynman ratio discriminates the reconstructing and nonreconstructing classes available on the finite clusters;
5. GFMC establishes the large-system momentum dependence of the corresponding spectral centroid up to 2048 sites.

Several statements remain beyond the calculation.

First, the sharpness of the reconstructed level is established only through 96 spins. GFMC determines ground-state quantities and therefore the spectral centroid, not the thermodynamic linewidth. The finite-size pole could broaden into a narrow continuum in the thermodynamic limit.

Second, Eq. (8) is the pure ring Hamiltonian in the charge-free ice manifold. Electric charges are absent by construction. Longer ring exchanges, virtual spinon processes, and other microscopic terms generated in a full spin Hamiltonian are outside the present calculation. The survival and quantitative position of the reconstructed branch under such perturbations remain open.

Third, the calculated observable is the sublattice-summed pseudospin response rather than the coherent material-specific neutron cross section. The experimental visibility of the low branch should therefore be tested by applying the appropriate magnetic projection for a chosen material.

ADDITIONAL SPECTRA

We thank the Digital Research Alliance of Canada for computational resources. The complete set of scripts generating every number, numerical gate, and figure reported in this work is included with the submission.

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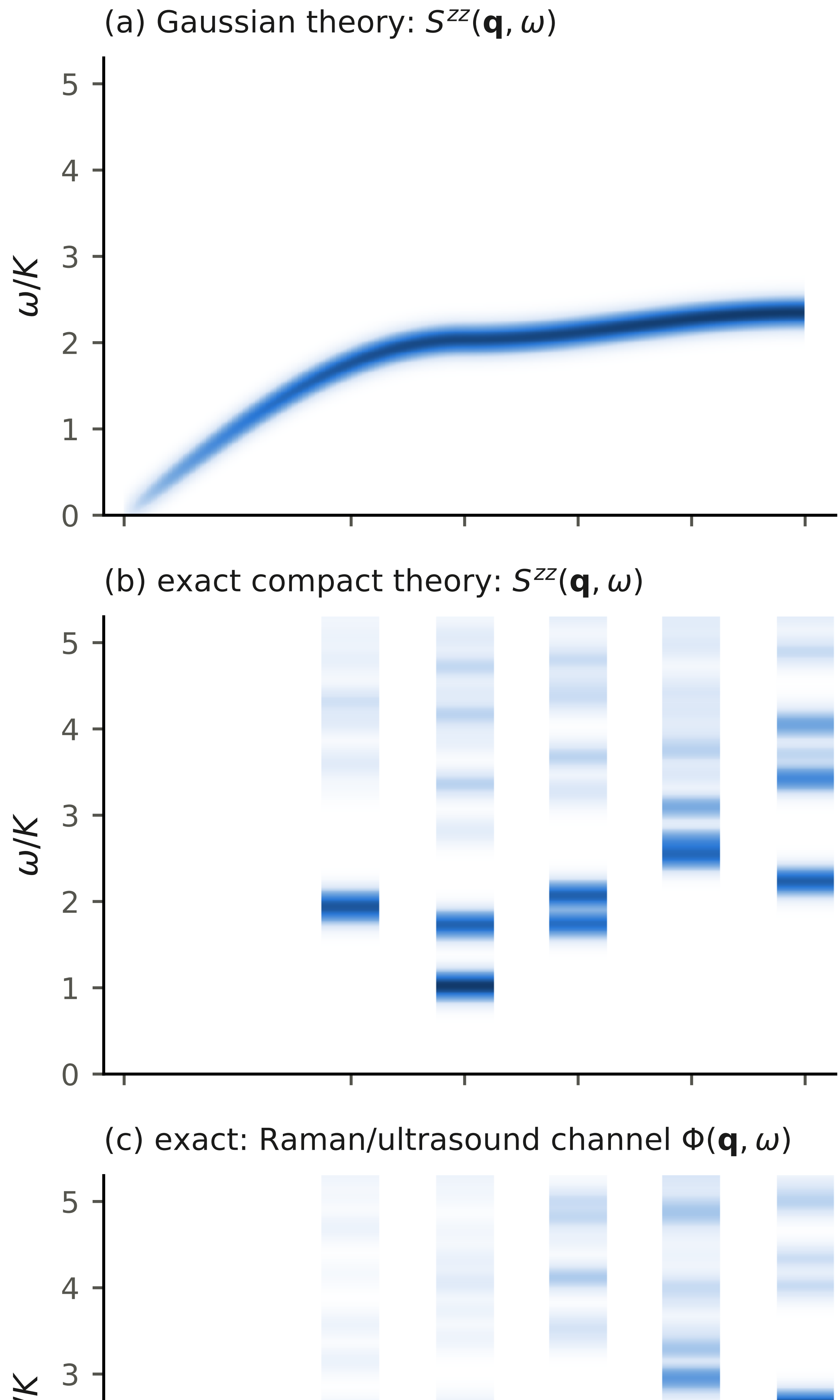
[2] E. Z. Simon, A. S. Patri, and Y. B. Kim, Phys. Rev. B **106**, 064427 (2022).
[3] G. K. Naik, J. N. Hallén, N. C. Jayarama, R. Moessner, and C. R. Laumann, Phys. Rev. Lett. (2026), arXiv:2512.14843.


(a) Gaussian theory: $S^{zz}(\mathbf{q}, \omega)$
ω/K
0
1
2
3
4
5
(b) exact compact theory: $S^{zz}(\mathbf{q}, \omega)$
ω/K
0
1
2
3
4
5
(c) exact: Raman/ultrasound channel $\Phi(\mathbf{q}, \omega)$
3
4
5

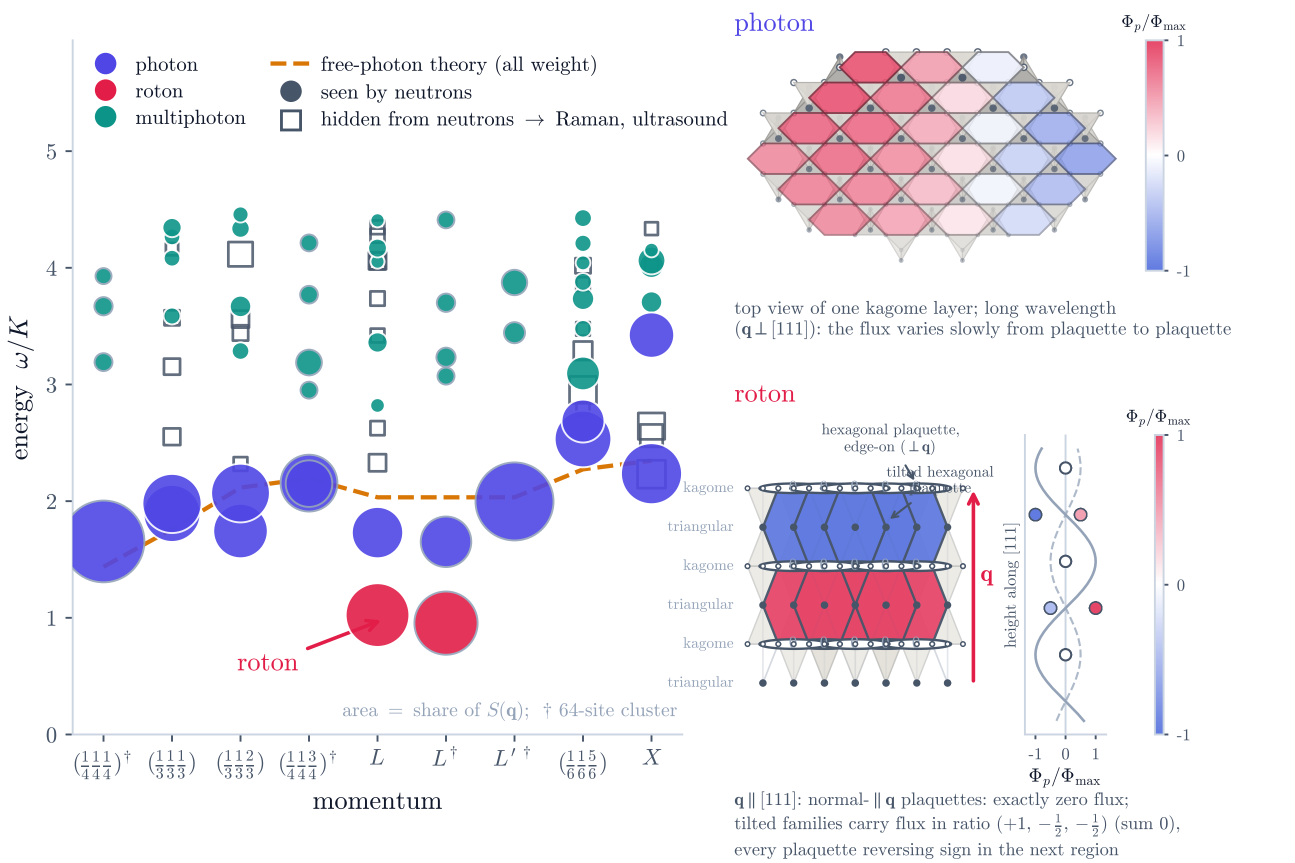
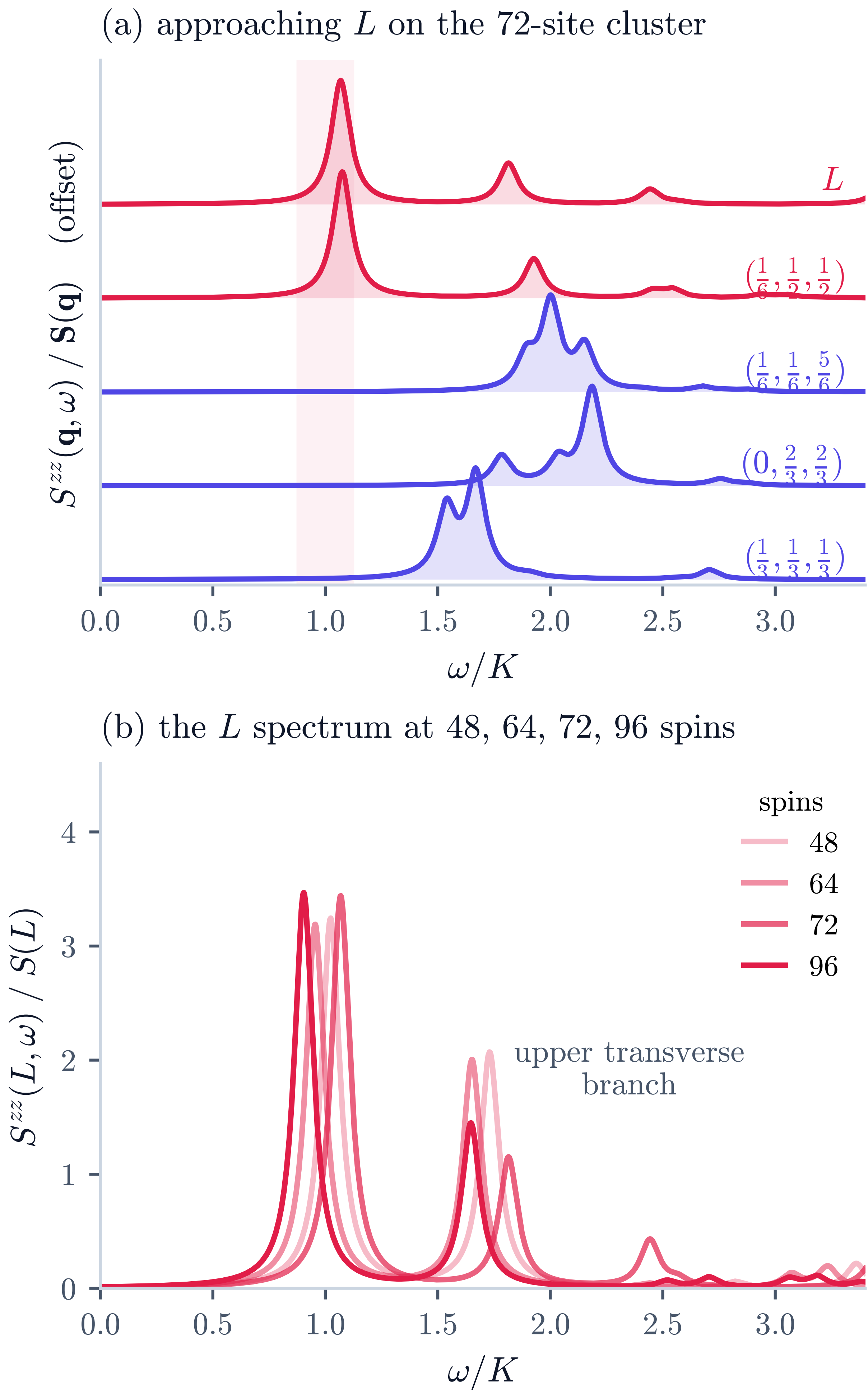


FIG. 2. Full operator-resolved finite-size spectrum and state diagnostics. Marker area represents the sublattice-summed S^z spectral weight.



(a) approaching L on the 72-site cluster
$S^{zz}(\mathbf{q},\omega)$ / $\mathbf{S}(\mathbf{q})$ (offset)
L
$(\frac{1}{6},\frac{1}{2},\frac{1}{2})$
$(\frac{1}{6},\frac{1}{6},\frac{5}{6})$
$(0,\frac{2}{3},\frac{2}{3})$
$(\frac{1}{3},\frac{1}{3},\frac{1}{3})$
0.0 0.5 1.0 1.5 2.0 2.5 3.0
ω/K
(b) the L spectrum at 48, 64, 72, 96 spins
$S^{zz}(L,\omega)$ / $S(L)$
spins
48
64
72
96
upper transverse branch
0 1 2 3 4
0.0 0.5 1.0 1.5 2.0 2.5 3.0
ω/K

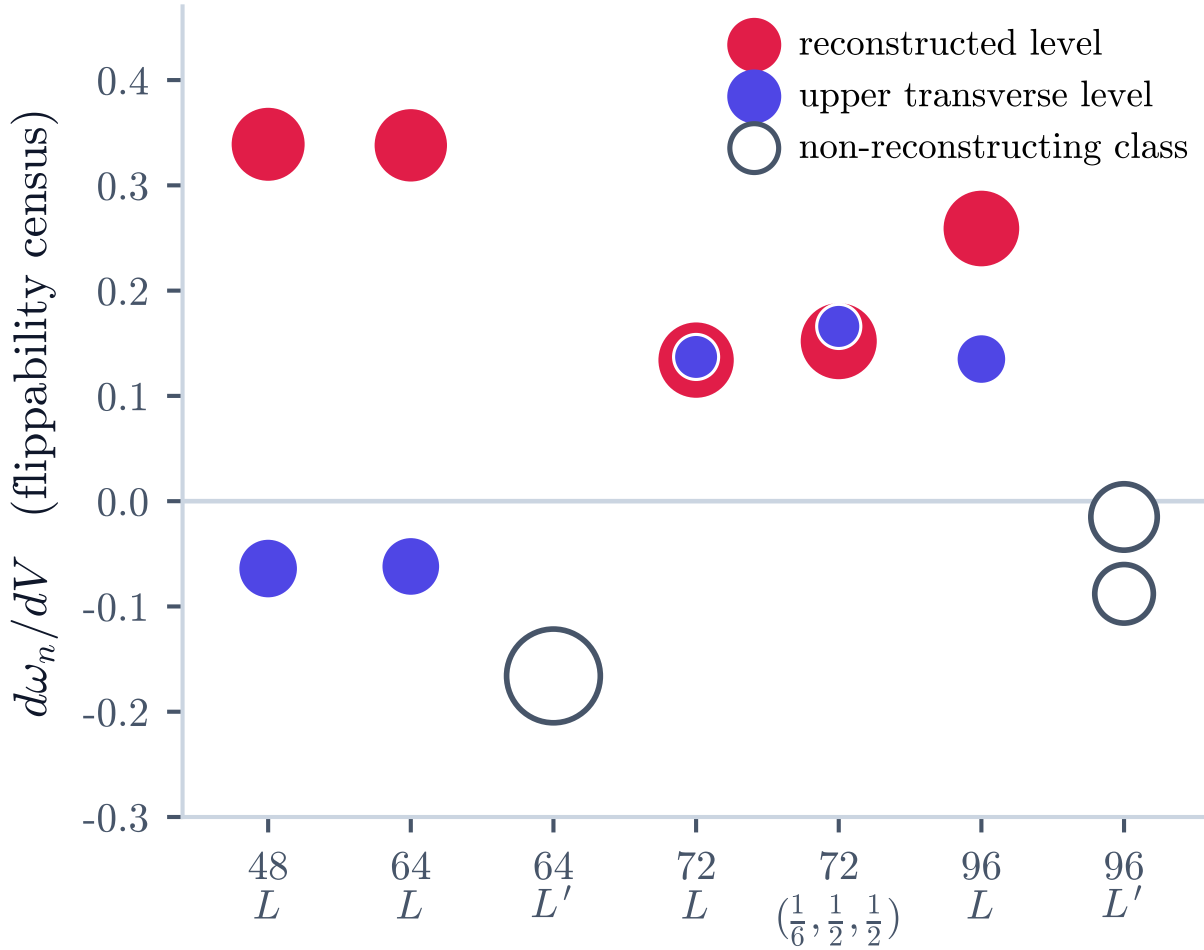


FIG. 4. Full RK deformation and flippability diagnostics for the finite-size states.

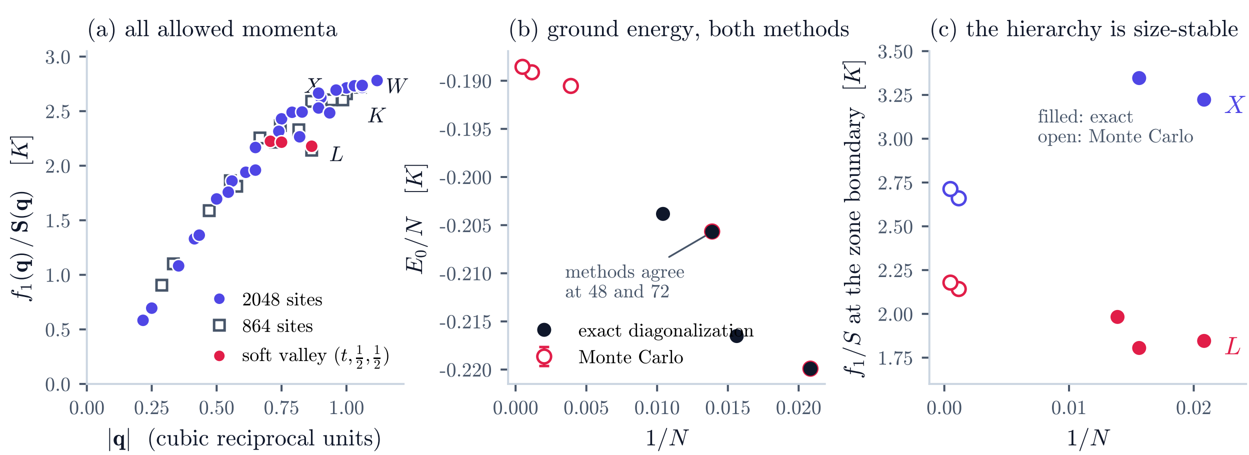


FIG. 5. Complete large-system GFMC diagnostics and Feynman-ratio landscape.